



[SIN-WI-2141_D] Work Instruction: Main Board and Ring Board (PCB) Acceptance Procedures

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	Effective date: 2021-11-23
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	Approval Date: 2025-11-04

Revision History

Revision	Changes	Author	Date
A	Initial Release	Tenzin Yangzom	2021-11-23

B	Updated section 3.0 - Training required to give more detail about the training required instead of just "YES". Updated section 1.0 - Introduction and added, "The Sinatra PCB is a component in the Sinatra device and incoming lots must undergo acceptance testing before moving the component into production." Added, "This work instruction is only valid for the PCB that is on version 0.1.4." to the scope - section 1.1. Added the callout for SIN-FRM-2206 from SINXXXXX when telling the operator to copy a form Added the EQUIP ID to the calliper in SINP-0010	Trevor Folska-Fung	2023-04-27
C	1. Sampling Plan: Section 4.1 and 4.2 are updated to follow the new AQL Sampling plan. We've shifted from Special S(4) to General S(1) for increased sampling per lot. 2. Test Record Form: Updated Sinatra Master Component Acceptance Test Record Form [SIN-FRM-2206_G] in SIN-001-001 PCB (Test ID: SINAT-0020) to ensure Micro-USB ports are checked for metal artifacts or debris for all the samples. 3. Work Instruction: Revised Sinatra Assembly Work Instruction [SIN-WI-2721_G]: ◦ Initial Stage Check: Verify that the Ray 1 charging cable inserts correctly into the socket during the initial staging to catch issues early.	Sahil Mathur	2024-08-06

	◦ Debris Prevention: Inspect Micro-USB ports for metal artifacts or debris before assembly.		
D	Revised Dimensional Inspection tests to reflect use of caliper for measurements. Removed references to previous testing involving other components. Added clarifications to the Physical Inspection steps and added step to test charging port on the board using a micro-USB cable. Added links to reference documents.	Rollin Luo	2025-11-04

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1. Introduction

The Sinatra PCB is a component in the Sinatra device and incoming lots must undergo acceptance testing before moving the component into production. This work instruction is to be used for acceptance testing and incoming inspection of the Sinatra Main Board and Ring Board (PCB).

1.1. Scope

This work instruction applies to the acceptance of the Sinatra Driver Board and Ring Board intended to be used with the Sinatra (Ray 1) system. This work instruction is only valid for the Printed Circuit Boards (PCB) that is on version 0.1.4.

2. Applicable and Reference Documents

2.1. Applicable Standards

Document Title	Document #
ANSI/ASQ Z1.4-2003	EXT-STD-0075

2.2. Reference Documents

Document Title	Document #
Sinatra Main Board and Ring Board (PCB) Procurement Specification	SIN-CS-2135
Sinatra PCB Drawing – Driver Board	Driverboard Assembly Drawing_v0.1.4.pdf
Sinatra PCB Drawing – Ring Board	Ringboard Assembly Drawing_v0.1.4.pdf
Device Integrated Verification and Validation Plan	SIN-PLN-2125
Sinatra Master Component Acceptance Test Record	SIN-FRM-2206

3. Training Required

YES - all persons performing acceptance testing on the Sinatra PCBs are to be trained prior to carrying out these procedures.

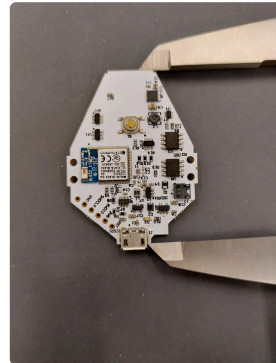
4. Work Instructions

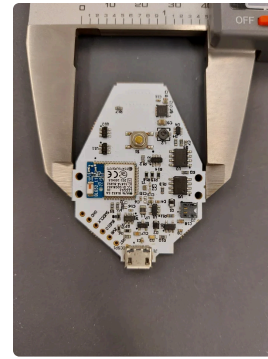
Complete the work instructions below and document in a copy of Sinatra Master Component Acceptance Test Record (SIN-FRM-2206) under the PCB tab. Upon completion, enter the document as a new record (REC type) in the Document Management System.

Inspection of the incoming Sinatra Driver Board and Ring Board (PCB) shall be performed prior to production. For each lot, an inspection lot shall be sampled throughout the received lot as per the Device Integrated Verification and Validation plan [SIN-PLN-2125].

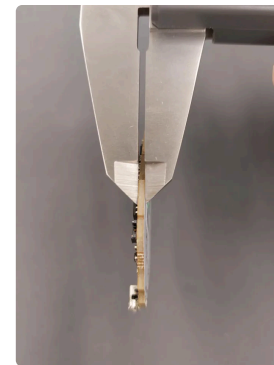
4.1. Dimensional Inspection

Test ID	SINP-0010	Test Name	Main Board and Ring Board (PCB) Dimensional Inspection
Traceability	PCB-DIM-0010 PCB-DIM-0020 PCB-DIM-0030 PCB-DIM-0040	Tools and Items Required	Caliper [EQUIP-001]

Step #	Task / Sub-tasks	Description
1	Collect PCBs for the Lot Under Test according to General - Inspection level I - AQL Level 4.0 sampling plan	
2	Using the calipers, measure the length, width, and thickness of each driver board sample. The expected range of dimensions for each are listed in SIN-FRM-2206 for PCB's.	 Length measurement



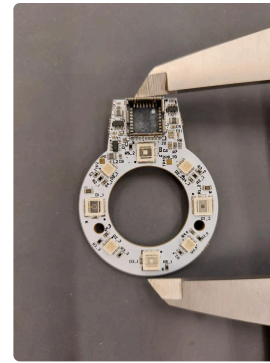
Width measurement



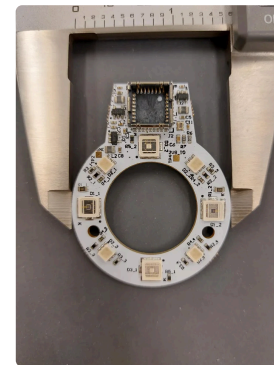
Thickness measurement

3

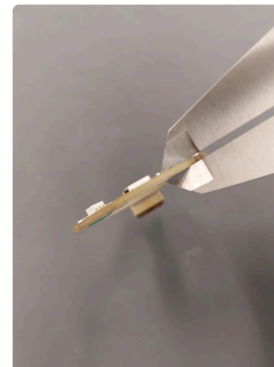
Using the calipers, measure the length, width, and thickness of each ring board sample. The expected range of dimensions for each are listed in SIN-FRM-2206 for PCB's.



Length measurement



Width measurement



Thickness measurement

4.2. Physical Inspection (Including Colour and Finish)

Test ID	SINP-0020	Test Name	Main Board and Ring Board (PCB) Physical Inspection
Traceability	PCB-PHY-0010	Tools and Items Required	Micro-USB connector cable
	PCB-PHY-0020		
	PCB-PHY-0030		
	PCB-PHY-0040		
	PCB-PHY-0050		
	PCB-PHY-0060		
	PCB-PHY-0070		
	PCB-PHY-0080		
	PCB-PHY-0090		
	PCB-PHY-0120		
	PCB-CFR-0010		
	PCB-CFR-0020		
	PCB-CFR-0030		

Step #	Task / Sub-tasks	Description
1	Collect PCBs for the Lot Under Test according to General - Inspection level I - AQL Level 4.0 sampling plan	
2	Ensure the PCBs adhere to the Physical Requirements outlined in SIN-CS-2135.	The colour and finish information will be found in the document. Ensure the colour and finish match through visual

		inspection and other physical requirements match.
3	Inspect the overall form of the PCBs, ensuring there is no damage found.	No burrs should be present on the edge of the PCBs (should be smooth). No cracks should be present on the PCBs. The PCBs should not be bent.
4	Refer to the PCB assembly drawings and ensure that the samples match.	Visually inspect to see if any components are missing on the PCBs. IE: The component is present in the BOM but not present on the board. Refer to the assembly drawings for what components should be present. Look for any exposed pads on the board which may be a component that has not been placed on the board.
5	Acquire a cable with a micro-USB connector.	
6	Plug in the micro-USB cable fully into the driver board to ensure that there are no metal artifacts or debris blocking the port. Check the inside of the port if there are any difficulties with plugging in the cable.	

4.3. Biological Review of Design

Test ID	SINP-0030	Test Name	Main Board and Ring Board (PCB) Biological Review of Design
Traceability	PCB-BIO-0010	Tools and Items Required	N/A visual inspection and reviewing documentation

Step #	Task / Sub-tasks	Description
1	Review Sinatra Main Board and Ring Board (PCB) procurement specifications (SIN-WI-2135) to confirm that the PCBs meet biological requirements.	Only repeat if spec or procurement requirements change

4.4. Cleanliness and Sterilization Review of Design

Test ID	SINP-0040	Test Name	Main Board and Ring Board (PCB) Cleanliness Review of Design
Traceability	PCB-CSR-0010 PCB-CSR-0020	Tools and Items Required	N/A visual inspection and reviewing documentation

Step #	Task / Sub-tasks	Description
1	Review Sinatra Main Board and Ring Board (PCB) procurement specification (SIN-WI-2135) to confirm presence of requirements for cleanliness and free of unnecessary debris	Only repeat if spec or procurement requirements change

✓ Page Attachments